University of Trier
Software Engineering Group
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Information Visualization
Summer Semester 2026
Exercise Sheet 7

Group work: Please keep in mind that the exercises are meant to prepare you for the exam.
ChatBots: Use your own brain as much and as often as possible.

Exercise 1 (Implementation of Word Cloud)

Implement a word cloud in Java or Python. As a starting point for your implementation you should use the code provided in `exercise07_code.zip` in StudIP. It contains a data structure for boxes. The provided code computes for a given list of words with weights a list of boxes including information on the width and height required to draw a word using a randomly selected font and a font size selected based on the weight. The code also includes a method that checks for collisions with all boxes that have been placed so far.

Your task is to implement the placement strategy in the function `place_box()` in Python or the method `placeBox()` in Java. It uses the Archimedean Spiral to find a position for a box such that it does not collide with a box that has already been placed.

Archimedean Spiral: `https://en.wikipedia.org/wiki/Archimedean_spiral`